TANISHA THAPA

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EDUCATION

MONASH UNIVERSITY | Master of Data Science | Melbourne, AU| Expected: December 2026
Relevant courses: Statistical Data Modelling, Big Data, Data Wrangling, Data Visualization, Data Analysis for Semi-Structured Data
CENTRALE MEDITERRANEE UNIVERSITY (Exchange Program) | Diploma in Data Science | Nice, FR | August 2023- August 2024
Relevant courses: Machine Learning, Agile Methodologies, Project Management
MAHINDRA UNIVERSITY | Bachelor of Technology in Computation & Mathematics | Hyderabad, IN | August 2020- June 2024
Relevant courses: Deep Learning, Mathematics in Machine Learning, Design & Analysis of Algorithms, Probability and Statistics

SKILLS

- **Programming Languages**: Python, SQL, R, MATLAB, Bash
- **Data and ML**: NumPy, Pandas, Scikit-learn, SciPy, NLTK, Matplotlib, Seaborn, BeautifulSoup, PyTorch, TensorFlow, LangChain, Transformers, SpaCy, Ollama, ChromaDB, ggplot2
- **Methods**: Time Series Analysis, Lag Analysis, Causality Testing, Econometrics, Deep Learning, NLP, Generative AI, Multimodal AI
- **Tools**: Git, Docker, Jira, Apache Hadoop/Spark, LIWC, Microsoft Excel, Tableau

EXPERIENCE

AI Intern | **Kheops** | March 2024 - August 2024
- Built a Project Management Workflow Engine automating task assignment, leave approval, and deadline notifications for a 10-person startup, replacing fully manual administrative processes and reducing operational overhead
- Advanced the AUTOPHP LLM task management system by redesigning it with a state machine architecture (task creation → execution → evaluation), improving modularity and scalability
- Integrated open-source LLMs via Hugging Face API/SDK as a cost-effective replacement for proprietary models, broadening the company's AI capabilities
- Used Spacy NER for smart task prioritisation, flagging urgent tasks and sending automated 3-day deadline reminders

ML Engineer | VOLTA Medical | Video Anonymization Application for Volta Medical | January 2024 - March 2024
- Developed an end-to-end video anonymisation application for cardiac surgery monitoring videos, reducing manual anonymisation time from 6 - 7 hours to 2 hours per 3-hour video - a 3x speed improvement
- Engineered a hybrid text detection pipeline combining YOLO (trained on 10,000 custom augmented images) and EAST (F1: 1.0), with Tesseract OCR and Spacy NER for precise patient name filtering; achieved 0.90 recall post-NER
- Implemented scene detection and screen splitting to avoid per-frame processing, cutting execution time significantly while maintaining accuracy
- Packaged the final product as a user-friendly executable (Gooey + PyInstaller) for hospital deployment, requiring no Python knowledge from end users

ML Engineer | MANE Fragrances | Chatbot for MANE's Legal and IP Department | October 2023 – December 2023
- Engineered an AI chatbot for MANE's Legal & IP department to streamline employee access to extensive legal documentation, reducing time spent searching regulations and querying supervisors
- Built the NLP pipeline by fine-tuning a Sentence-BERT model on a manually augmented dataset, enabling accurate semantic search over legal documents
- Integrated Vicuna LLM as a fallback for low-confidence queries, and delivered the solution via a Streamlit web interface accessible to non-technical legal staff

KEY PROJECTS

F1 Global Fandom Growth: Interactive Data Visualisation | 2026 | Interactive Dashboard
- Engineered a multi-source data pipeline combining Google Trends, FastF1 API, and manually compiled event data (154+ country-year records); solved a Google Trends window-normalisation problem using overlapping-window scaling to produce one continuous comparable time series
- Applied OLS regression, paired t-tests, and correlation analysis to quantify Drive to Survive's impact on fan engagement ($t = -3.29$, $p = 0.006$, +42% post-release), isolating event-driven vs. performance-driven growth across 14 countries
- Designed and built an interactive D3.js visualisation following the Five Design Sheet methodology, translating Tableau exploratory findings into a narrative-driven public-facing tool

Behavioral Stress Index (BSI): Economic Forecasting | 2025
- Built an interpretable Behavioural Stress Index (BSI) from Australian Google Trends and ABS macro data using PCA and Ridge Regression, identified top behavioural drivers (Centrelink, mortgage stress, food bank) correlated with unemployment.
- Implemented a walk-forward stress classifier predicting forward unemployment spikes, achieving ROC AUC 0.71 with a median lead time of 8 months ahead of traditional indicators.